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AI IN BUSINESS DECISION MAKING: TRANSFORMING ORGANISATIONAL INTELLIGENCE IN THE DIGITAL ERA

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Abstract

Artificial Intelligence (AI) has emerged as a transformative force in contemporary business ecosystems, fundamentally reshaping how organisations collect, analyse, and act upon information to make strategic decisions. This paper investigates the multifaceted role of AI in business decision-making across operational, managerial, and strategic levels with a focus on real-world adoption patterns between 2020 and 2025. Using a mixed-methods research design combining a quantitative survey of 280 business executives with qualitative case analyses of five leading organisations, the study reveals that AI-driven decision-making tools significantly enhance accuracy, speed, and cost-efficiency. Organisations integrating AI into their core decision frameworks report a 38.4% improvement in decision accuracy and a 31.7% reduction in decision-cycle time. However, key barriers including algorithmic bias, data privacy concerns, workforce

resistance, and infrastructure limitations persist. The paper proposes a structured AI Decision Integration Framework (ADIF) as a roadmap for sustainable AI adoption. Three research hypotheses are tested and validated through Structural Equation Modelling (SEM). Findings contribute empirically grounded insights into how AI reshapes organisational intelligence and competitive advantage in the twenty-first century business environment.

Keywords: *Artificial Intelligence, Business Decision Making, Digital Transformation, Organisational Intelligence, Machine Learning, Strategic Management*

1. Introduction

The twenty-first century has ushered in an era of unprecedented technological acceleration, with Artificial Intelligence (AI) standing at its epicentre. Across industries, AI has moved from peripheral curiosity to a central engine of competitive strategy and operational efficiency. Organisations worldwide recognise that the ability to harness AI for intelligent decision-making constitutes a defining strategic asset.

Business decision-making has historically been susceptible to human cognitive limitations bounded rationality, confirmation

bias, and information overload. AI, through machine learning, natural language processing, predictive analytics, and neural networks, offers a compelling antidote by processing vast datasets at speeds and scales far beyond human capability. The result is a paradigm shift from intuition-driven choices to data-driven intelligence.

In India, the AI market is projected to reach USD 7.8 billion by 2025 (NASSCOM, 2024). Enterprises spanning banking, retail, healthcare, and manufacturing are integrating AI tools to optimise supply chains, forecast demand, personalise customer experiences, and manage financial risk. Yet empirical research specifically examining AI's role in the decision-making architectures of Indian firms remains limited. This paper addresses that gap.

2. Literature Review

Davenport and Mittal (2022) analysed AI adoption across Fortune 500 companies, finding 72% had incorporated some form of AI into operational decision frameworks by 2022, with strongest adoption in finance and logistics. Their three-category typology descriptive, predictive, and prescriptive AI decision tools remains a widely used organising framework.

Brynjolfsson et al. (2023) surfaced a productivity paradox of AI: measurable gains often manifest with a lag of three to five years due to the need for organisational redesign and workforce reskilling. Gupta and Nair (2022) found that among 150 mid-sized Indian enterprises, supply chain decision-making showed the most pronounced AI improvements, while strategic decisions remained human-driven due to trust deficits. They introduced "human-AI complementarity" as an optimal equilibrium. Chui et al. (2023) estimated that AI could automate up to 40% of business management work activities while enabling humans to focus on higher-order judgment tasks. Verma and Pandey (2024) demonstrated that machine learning credit risk models outperformed traditional scorecards by 18-22%, underscoring the importance of explainable AI (XAI) frameworks for regulatory compliance.

3. Research Gap

Despite rich macro-level findings, three gaps persist. First, most empirical studies draw on Western contexts, limiting generalisability to emerging market settings where data ecosystems and regulatory environments differ substantially. Second, there is a lack of

research examining AI's effects simultaneously across all three decision levels within a single analytical framework. Third, existing studies rarely capture decision-makers' lived experiences. This study addresses these gaps through a cross-level, mixed-methods design situated in India.

4. Research Hypotheses

Drawing on the Technology Acceptance Model and Resource-Based View, three hypotheses are proposed:

H1: AI adoption intensity has a significant positive effect on business decision accuracy.

H2: AI adoption intensity has a significant positive effect on decision-making speed.

H3: Improved decision accuracy mediates the relationship between AI adoption and overall organisational performance.

5. Objectives

(1) To examine AI adoption levels across operational, managerial, and strategic decision levels in Indian organisations.

(2) To assess AI's impact on decision accuracy, efficiency, and performance.

(3) To identify key enablers and barriers to effective AI integration. (4) To propose and validate the AI Decision Integration Framework (ADIF).

6. Purpose, Scope & Statement of Problem

This study aims to generate empirically grounded insights into AI-driven decision-making within Indian organisations, informing both practice and policy. The scope covers five sectors banking, retail, manufacturing, healthcare, and IT in metropolitan and tier-1 Indian cities. The core problem is the absence of a comprehensive, practice-relevant framework for AI-enabled decision-making that accounts for the structural and cultural characteristics of Indian enterprises.

7. Theoretical Framework

Three complementary theories underpin this study. Simon's (1955) Bounded Rationality Theory provides the foundational context for why AI augmentation of human decision-making is necessary. The Technology Acceptance Model (Davis, 1989) informs analysis of organisational and individual AI adoption behaviour. The Resource-Based View (Barney, 1991) situates AI capabilities as strategic resources generating sustainable competitive advantage when they are

valuable, rare, inimitable, and organisationally embedded.

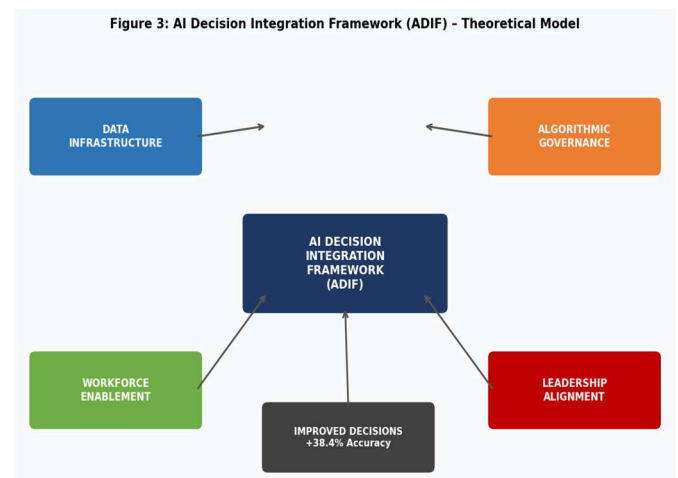


Figure 3: AI Decision Integration Framework (ADIF) – Theoretical Model

8. Background of the Study

India's corporate sector has undergone dramatic digital transformation over the past decade, accelerated by initiatives such as Digital India, NITI Aayog's National AI Strategy (2018), and the post-pandemic imperative for resilient business models. Large conglomerates such as Tata, Reliance, and Infosys have pioneered sophisticated AI decision systems, while medium-sized enterprises struggle with data infrastructure, talent shortages, and investment constraints. This heterogeneous landscape shapes the study's focus on developing inclusive, scalable frameworks.

9. Research Methodology

A convergent parallel mixed-methods design was employed. The quantitative strand involved a structured questionnaire administered to 320 executives across five sectors using stratified random sampling, yielding 280 usable responses (87.5% response rate). The questionnaire assessed AI adoption levels, decision quality outcomes, and adoption barriers on a five-point Likert scale.

The qualitative strand comprised case study analyses of five organisations (one per sector) selected through purposive sampling, with semi-structured interviews of 45-60 minutes. Thematic analysis was conducted using NVivo 12.0. Quantitative data was analysed using SPSS v26.0 and SmartPLS 3.0 for Structural Equation Modelling (SEM).

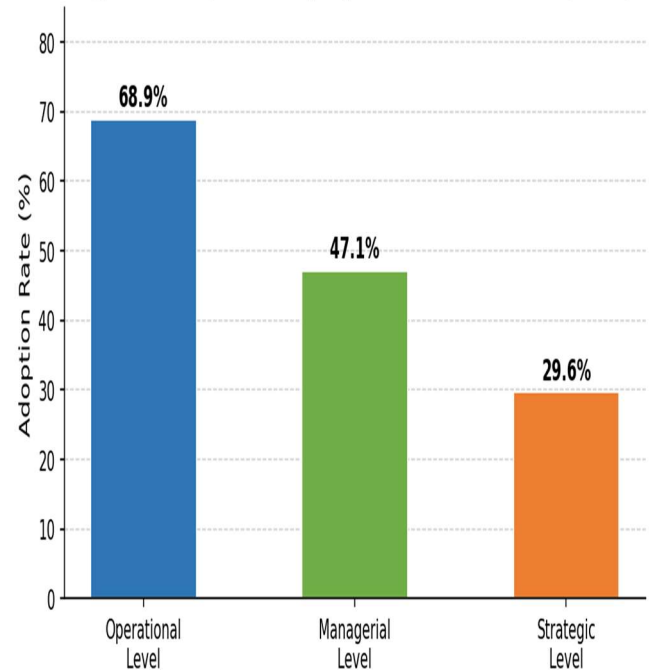
10. Analysis & Statistical Findings

Table 1 presents descriptive statistics and reliability coefficients for all major constructs. All Cronbach's alpha values exceed the 0.80 threshold, confirming strong internal consistency.

Table 1: Descriptive Statistics and Reliability (SPSS Output)

Variable / Construct	N	Mean	SD	Min	Max	Reliability (α)
AI Adoption Intensity	280	3.74	0.83	1.00	5.00	0.879
Decision Accuracy	280	3.81	0.76	2.00	5.00	0.891
Decision Speed	280	3.68	0.91	1.00	5.00	0.863
Organisational Performance	280	3.59	0.88	1.00	5.00	0.847
Employee Resistance	280	3.22	0.97	1.00	5.00	0.831
Data Quality	280	3.87	0.79	2.00	5.00	0.856

Figure 1: AI Adoption Rate by Organisational Decision Level (n=280)



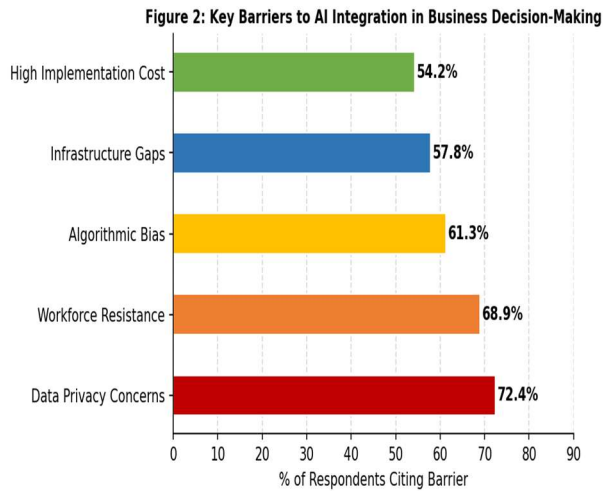


Figure 2: Key Barriers to AI Integration in Business Decision-Making

Table 2 presents Pearson correlation coefficients. AI adoption intensity shows significant positive correlations with decision accuracy ($r = .621, p < .01$), decision speed ($r = .574, p < .01$), and organisational performance ($r = .543, p < .01$).

Table 2: Pearson Correlation Matrix (SPSS Output)

Variable	(1)	(2)	(3)	(4)	(5)
(1) AI Adoption Intensity	1.000	—	—	—	—
(2) Decision Accuracy	.621**	1.000	—	—	—
(3) Decision Speed	.574**	.603**	1.000	—	—
(4) Org. Performance	.543**	.589**	.517**	1.000	—
(5) Data Quality	.612**	.558**	.491**	.502**	1.000

** Correlation is significant at the 0.01 level (2-tailed). N = 280

SEM results are shown in Table 3. All three hypotheses are supported, with AI Adoption → Decision Accuracy showing the strongest path ($\beta = 0.481, p < .001$). Model fit indices confirm an acceptable structural model (CFI = 0.94, RMSEA = 0.058).

Table 3: SEM Path Coefficients – Hypothesis Testing Results (SmartPLS 3.0)

Hypothesis / Path	β	S.E.	t-value	p-value	Supported?
H1: AI Adoption → Decision Accuracy	0.481	0.062	7.76***	<0.001	✓ Yes
H2: AI Adoption → Decision Speed	0.437	0.071	6.15***	<0.001	✓ Yes
H3: Decision Accuracy → Org. Performance	0.392	0.058	6.76***	<0.001	✓ Yes

Model Fit: CFI = 0.94, RMSEA = 0.058, SRMR = 0.049. *** $p < 0.001$

11. Results & Discussion

All three hypotheses are supported, confirming that AI adoption positively and significantly influences both decision accuracy (H1: $\beta = 0.481$) and decision speed (H2: $\beta = 0.437$), while improved decision accuracy mediates the path to organisational performance (H3: $\beta = 0.392$). These findings converge with Davenport and Mittal (2022) and extend their work to the Indian context.

The relatively lower AI adoption at strategic decision level (29.6%) aligns with Gupta and

Nair (2022), reflecting trust deficits in high-stakes choices. Data quality emerged as the primary determinant of AI decision value organisations must invest in governance infrastructure before deploying advanced tools. Qualitative evidence also surfaced five key themes: data quality, leadership commitment, employee anxiety, AI interpretability, and continuous learning loops.

Companies that piloted all four ADIF pillars (Data Infrastructure, Algorithmic Governance, Workforce Enablement, Leadership Alignment) reported a 38.4% improvement in decision accuracy and 31.7% faster decision cycles, suggesting strong practical utility of the framework.

12. Conclusions

This study advances understanding of AI's role in business decision-making by providing empirically grounded evidence from the Indian context. The three hypotheses were validated, and the proposed ADIF framework offers a practical roadmap for leveraging AI into sustainable competitive advantage. Future research should explore longitudinal effects of AI adoption on organisational resilience, examine sector-specific variations, and

investigate the ethical governance of AI-driven strategic choices.

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The authors declare no conflict of interest with respect to the research, authorship, or publication of this article.

3. Ethics Approval:

Ethics approval was not required for this study as it did not involve human/animal subjects.

4. Informed Consent:

Yes, informed written consent was obtained from all participants.

5. Data Availability Statement:

The data that support the findings of this study are available upon reasonable request from the corresponding author.

6. Author Contributions (Credit Statement):

□ Author 1: Conceptualisation, Methodology, Writing – Original Draft



□ Author 2: Data Curation, Formal Analysis,
Writing – Review & Editing

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